

PROJECT PROPOSAL

CS3631 – Deep Neural Networks · Research Project (2026) · Group 04

Trustworthy Deep Learning for Chest X-ray Disease Detection

A model-agnostic Shortcut-Suppression Module, evaluated through a five-axis trustworthiness benchmark across CNN and Transformer families.

|                    |                                                                               |
|--------------------|-------------------------------------------------------------------------------|
| Research objective | Improve an existing DNN (§3-2): a novel module that reduces shortcut learning |
| Core contribution  | Model-agnostic Shortcut-Suppression Module (3 candidate designs)              |
| Evaluation         | Five-axis trustworthiness benchmark across 4 architectures                    |
| Baseline (§4)      | DenseNet121 (CNN), ImageNet-pretrained                                        |
| Architectures      | ResNet50 · DenseNet121 · EfficientNet-B0 · ViT-Base                           |
| Datasets           | COVID-19 Radiography (primary) · RSNA (external OOD)                          |
| Compute            | Free Google Colab / Kaggle GPU (§6)                                           |
| Team / Duration    | 5 members · one month                                                         |

# Contents

## 1. Abstract

Deep-learning chest X-ray (CXR) classifiers report accuracies above 95%, but much of that performance comes from shortcut learning — exploiting hospital-source markers or acquisition artefacts rather than lung pathology — so accuracy collapses on data from unseen hospitals (AUC falling from  $>0.99$  to  $\sim 0.76$ ). A clinically usable CXR model must additionally attend to the diseased lung, be well-calibrated, and generalise across hospitals; yet these properties are studied in isolation, and no lightweight architectural fix for shortcut learning has been shown to generalise across model families.

This project makes a two-fold contribution: (1) a novel, model-agnostic **Shortcut-Suppression Module** — investigated as three candidate designs (a lung-region attention block, an auxiliary segmentation head, and a Grad-CAM-based suppression loss) — that forces any backbone to attend to clinically relevant lung regions; and (2) a **five-axis trustworthiness benchmark** (accuracy, calibration, explanation faithfulness, efficiency, cross-dataset robustness) that evaluates the module in a controlled, before-and-after manner. The winning module is applied to four architectures (ResNet50, DenseNet121, EfficientNet-B0, ViT-Base), turning the multi-architecture sweep into a generalisation test of the fix across CNN and Transformer families.

## 2. Problem Motivation

Chest X-ray is the most common medical imaging examination worldwide, and automated screening is especially valuable where radiologists are scarce. But high headline accuracy is misleading: models frequently exploit dataset artefacts — hospital source markers, acquisition-device signatures, text and borders — rather than genuine lung pathology. When tested on data from unseen hospitals, performance drops sharply (AUC from above 0.99 to around 0.76). A CXR model intended for clinical use must therefore satisfy three conditions beyond accuracy:

1. It must focus on the diseased lung region rather than image artefacts.
2. Its confidence scores must be calibrated so clinicians can trust predicted probabilities.
3. It must generalise to unseen hospital data.

Existing studies address these properties separately. No work combines accuracy, calibration, explanation faithfulness, computational efficiency and cross-dataset robustness into one controlled comparison, nor proposes a lightweight architectural fix for shortcut learning that generalises across model families. This project addresses both gaps.

## 3. Research Gap & Positioning

Two gaps define the opportunity, and our contribution targets both.

- **No cross-family shortcut fix.** Prior debiasing methods are typically demonstrated on a single backbone. Whether a lightweight module can suppress shortcut learning across both CNNs and Vision Transformers is unknown.
- **No unified trustworthiness evaluation.** Accuracy, calibration, faithfulness, efficiency and robustness are rarely measured together under one identical protocol, so it is unclear whether a fix genuinely reduces shortcut reliance or merely shifts the problem.

### Our contribution answers both

We propose a model-agnostic Shortcut-Suppression Module and prove it generalises by plugging it into four architectures; and we build a five-axis trustworthiness benchmark that measures, before and after, whether it truly reduces shortcut reliance.

This is a genuine DNN contribution (a novel module / learning strategy), not a benchmark of pretrained models — the benchmark is the module's evaluation framework.

## 4. Identified Datasets

| Dataset                                | Content                                                                      | Role                                                                                                                |
|----------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| COVID-19 Radiography Database (Kaggle) | ~21,000 images · 4 classes (COVID-19, Normal, Viral Pneumonia, Lung Opacity) | PRIMARY — train / val / test. Known source bias makes it ideal for studying shortcut learning.                      |
| RSNA Pneumonia Detection (Kaggle)      | Frontal CXRs (Pneumonia / Normal) from different hospital sources            | EXTERNAL — out-of-distribution robustness only; never trained on. Internal→external drop is the robustness measure. |

Both are permitted under (Kaggle / research benchmark datasets). The class imbalance in the primary set is handled with class weighting.

## 5. Baseline Model (Mandatory, §4)

Since the data are images, the baseline is a CNN. We select **DenseNet121** as the primary baseline for its established medical-imaging performance via dense feature reuse (~8.0 M parameters), ImageNet-pretrained, with the final layer replaced by a 4-class head.

### 5.1 Implementation details

- Images resized to 224×224, converted to 3-channel, normalised with ImageNet statistics.
- Stratified 70/15/15 train/val/test split, fixed seed.
- Augmentation: random rotation, horizontal flip, brightness/contrast jitter, random cropping.
- PyTorch + timm; AdamW; cross-entropy with class weighting; Google Colab / Kaggle free GPUs.

### 5.2 Hyper-parameter tuning (mandatory, §12)

Default hyper-parameters give limited accuracy, so we conduct systematic tuning over learning rate, batch size, optimiser (Adam, AdamW, SGD+momentum), weight decay, and LR schedulers (cosine annealing, step decay, reduce-on-plateau). We use a two-phase transfer-learning strategy — Phase 1 freezes the backbone and trains the head; Phase 2 unfreezes the final dense blocks for fine-tuning at a reduced learning rate — with early stopping on validation loss. Tuning is applied identically across all models for fair comparison.

## 6. Proposed Contribution

Our contribution is twofold: a novel lightweight architectural addition to suppress shortcut learning, and a unified multi-axis trustworthiness benchmark to evaluate it.

### 6.1 Shortcut-Suppression Module (three candidate designs)

A lightweight, model-agnostic module that can be inserted into any backbone to force the network to attend to clinically relevant lung regions rather than dataset artefacts. We investigate three candidates:

| Candidate                        | Mechanism                                                                                                                                                                                     | Notes                          |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| (a) Lung-Region Attention Module | A spatial attention block guided by an automatically generated lung mask that re-weights feature maps to suppress background and artefact regions.                                            | Low risk; primary candidate    |
| (b) Auxiliary Segmentation Head  | A secondary decoder branch on intermediate features that jointly predicts a lung mask during training, acting as an anatomical regulariser steering the shared backbone toward lung features. | Medium risk; primary candidate |
| (c) Shortcut-Suppression Loss    | A custom regularisation loss penalising the model when Grad-CAM activations fall outside the lung region — directly optimising explanation faithfulness during training.                      | Higher risk; exploratory       |

Each candidate is developed and ablated against the vanilla DenseNet121 baseline to isolate its effect. The winning design is then plugged into all four architectures to demonstrate that the fix generalises across CNN and Transformer families, not just one model.

### 6.2 Five-Axis Trustworthiness Benchmark

The multi-architecture comparison is the evaluation framework for the proposed module. We benchmark all models — with and without the module — across five axes:

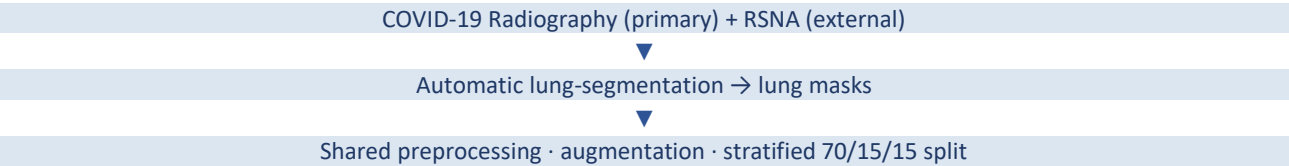
| Axis                     | Measured by                                                                                            |
|--------------------------|--------------------------------------------------------------------------------------------------------|
| Accuracy                 | Accuracy, precision, recall, F1, ROC-AUC (per-class + macro), confusion matrix                         |
| Calibration              | Expected Calibration Error (ECE), Brier score, reliability diagrams — before/after temperature scaling |
| Explanation faithfulness | Grad-CAM (CNNs), attention rollout (ViT), and a quantitative lung-localisation score                   |
| Efficiency               | FLOPs, parameters, model size, inference latency (CPU/GPU)                                             |
| Robustness               | Internal→external (out-of-distribution) accuracy & AUC drop                                            |

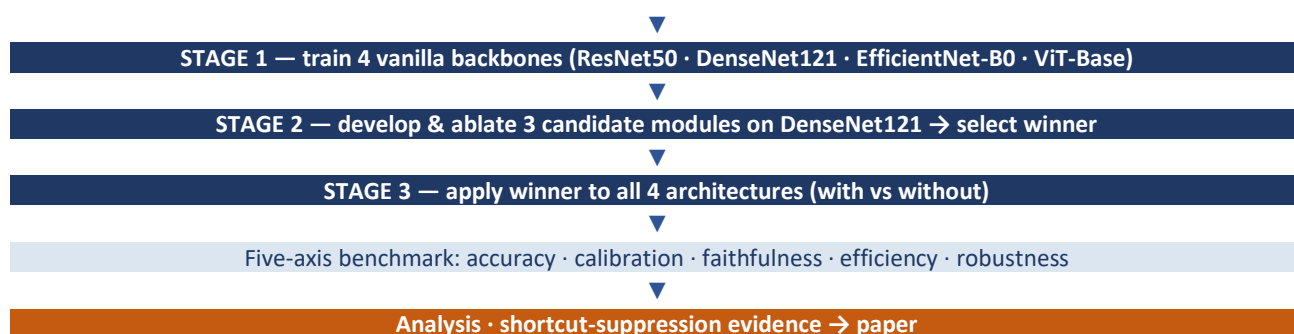
**Why the design is strong**

The controlled before-and-after comparison across architectures produces strong evidence of whether the module genuinely reduces shortcut reliance or merely shifts the problem — and the cross-architecture sweep doubles as a generalisation test of the contribution.

## 7. Methodology & Pipeline

Lung masks (needed by all three candidates and the faithfulness score) are produced by an automatic lung-segmentation model. All backbones share one identical preprocessing/training pipeline so any difference is attributable to the architecture and the module.





## 8. Experimental Plan

### 8.1 Three stages

| Stage                                       | What happens                                                                                                                                                                | Output                     |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| Stage 1 — Baseline establishment            | Train all four vanilla architectures under an identical pipeline (same preprocessing, augmentation, optimiser, split, seed); frozen vs fine-tuned; report baseline metrics. | Baseline table (5 axes)    |
| Stage 2 — Module development & ablation     | Develop the three candidate modules on DenseNet121; ablate each component; select the best by faithfulness + robustness gain without significant accuracy loss.             | Selected module + ablation |
| Stage 3 — Cross-architecture generalisation | Apply the selected module to all four architectures; compare with-module vs without-module across all five axes.                                                            | Generalisation evidence    |

### 8.2 Evaluation metrics

Classification: accuracy, precision, recall, F1, ROC-AUC (per-class + macro), confusion matrices. Calibration: ECE, Brier, reliability diagrams before/after temperature scaling. Explainability: heatmap comparison + quantitative lung-localisation faithfulness score. Robustness: internal-to-external accuracy & AUC drop. Efficiency: parameters, FLOPs, model size, inference time (CPU & GPU).

## 9. Feasibility & Scope Management (One Month)

The plan is ambitious but achievable in one month with a five-member team, provided three risks are actively managed.

| Risk to the timeline                                                            | How we keep it feasible                                                                                                                                           |
|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lung masks are on the critical path (needed by all 3 candidates + faithfulness) | Use an off-the-shelf / pretrained lung-segmentation model (or a quickly-trained U-Net on public lung-mask data) in Week 1 — do not sink a week into segmentation. |
| Candidate (c) Grad-CAM-in-the-loss is technically hard and slow                 | Treat (a) and (b) as primary candidates and (c) as exploratory; the winner can be selected from (a)/(b) if (c) does not converge in time.                         |
| Stage 3 is a large matrix (4 archs × with/without × 5 axes)                     | Build the benchmark harness once (reusable); most of Stage 3 is then running jobs, parallelised across members.                                                   |
| ViT-Base training cost                                                          | 224×224, mixed precision, ImageNet-pretrained init; DeiT/ViT-Small fallback if unstable.                                                                          |

#### Feasibility verdict

Feasible in 4 weeks IF: (1) lung masks come from an existing model, (2) candidate (c) is de-risked to exploratory, and (3) the 5 members work in parallel — each owning one architecture plus one module/axis. The riskiest single item is candidate (c); the project does not depend on it.

## 10. Computational Feasibility & Rules Compliance

All experiments run on free Google Colab / Kaggle GPUs at 224×224 with mixed-precision training; no multi-GPU clusters or HPC are required. Frequent checkpointing guards against session limits.

| Rule                 | Compliance                                                                                                                               |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| §2 Dataset           | COVID-19 Radiography + RSNA (Kaggle) — permitted under §2.2.                                                                             |
| §3 Objective         | Objective 2 — improve an existing DNN via a novel shortcut-suppression module.                                                           |
| §4 Baseline          | DenseNet121 CNN baseline, fully evaluated.                                                                                               |
| §5 Model constraints | CNNs + ViT + a lightweight novel module — the contribution is the module, not merely applying pretrained models. No LLM/agentic methods. |
| §6 Compute           | Runs on free Colab/Kaggle GPU.                                                                                                           |
| §8 Contribution      | Novel architecture / learning strategy (the module) + explainability framework.                                                          |
| §9 Experiments       | Baseline comparison, HP tuning, ablation studies, computational analysis — all included.                                                 |
| §11 Individual work  | Every member owns a hands-on DNN task — a model and a module/axis (see §12).                                                             |

## 11. Timeline (One Month)

| Week | Focus                                                                       | Output                                    |
|------|-----------------------------------------------------------------------------|-------------------------------------------|
| 1    | Pipeline + lung masks; Stage 1 — train 4 vanilla backbones + HP tuning      | Baseline metrics (feeds the ACM proposal) |
| 2    | Stage 2 — develop 3 candidate modules on DenseNet121; ablate; select winner | Selected module + ablation                |
| 3    | Stage 3 — apply winner to all 4 archs; run 5-axis benchmark (with/without)  | Full results + figures                    |
| 4    | Consolidate, statistical analysis, write paper + slides                     | Conference-ready paper + code             |

## 12. Team Roles (§11 — each owns a DNN model + a module/axis)

Every member trains at least one model and owns part of the novel contribution, so each makes an identifiable DNN-development contribution — not merely writing.

| Member | Architecture (trains)                         | Module / axis ownership                                                        | Paper section                 |
|--------|-----------------------------------------------|--------------------------------------------------------------------------------|-------------------------------|
| M1     | ResNet50                                      | Candidate (a) Lung-Region Attention Module                                     | Intro, Related Work           |
| M2     | DenseNet121 (primary baseline)                | Data pipeline + lung-mask generation + HP-tuning standard                      | Baseline, Experimental Setup  |
| M3     | EfficientNet-B0                               | Candidate (b) Auxiliary Segmentation Head                                      | Proposed Module               |
| M4     | ViT-Base                                      | Candidate (c) Shortcut-Suppression Loss + attention-rollout faithfulness       | Method (loss) + Discussion    |
| M5     | Lung-segmentation model + winner ×4 (Stage 3) | Five-axis benchmark harness: calibration, robustness, faithfulness, efficiency | Results, Conclusion, assembly |

## 13. Risks & Mitigations

| Risk                                                        | Mitigation                                                                                                                   |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| A candidate module fails to improve faithfulness/robustness | Three candidates de-risk the contribution; report the negative result via the controlled ablation and proceed with the best. |
| Grad-CAM-based loss (c) does not converge                   | Keep (a)/(b) as primaries; (c) is exploratory and non-blocking.                                                              |

| Risk                                 | Mitigation                                                                                                               |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Lung masks imperfect                 | Use masks as soft guidance; the faithfulness score tolerates approximate masks; fall back to an off-the-shelf segmenter. |
| ViT over-fits on the smaller dataset | ImageNet-pretrained init, strong augmentation, DeiT/ViT-Small fallback.                                                  |
| Free-GPU limits                      | Mixed precision, checkpointing, staggered jobs across accounts.                                                          |

## 14. Expected Contributions

4. A novel, model-agnostic Shortcut-Suppression Module that reduces reliance on dataset artefacts in CXR classification.
5. Evidence that the fix generalises across CNN and Transformer families (applied to four architectures).
6. A unified five-axis trustworthiness benchmark (accuracy, calibration, faithfulness, efficiency, robustness) with controlled before/after comparison.
7. A quantitative lung-localisation faithfulness score linking interpretability to robustness.
8. Reproducible, free-hardware code and practical model-selection guidance for deployment.

## 15. Indicative References

- [1] COVID-19 Radiography Database, Kaggle. [kaggle.com/datasets/tawsifurrahman/covid19-radiography-database](https://kaggle.com/datasets/tawsifurrahman/covid19-radiography-database).
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- [3] Huang et al. Densely Connected Convolutional Networks (DenseNet). CVPR 2017.
- [4] Ronneberger et al. U-Net: Convolutional Networks for Biomedical Image Segmentation. MICCAI 2015.
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- [6] Geirhos et al. Shortcut Learning in Deep Neural Networks. Nature Machine Intelligence, 2020.
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- [9] Dosovitskiy et al. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale (ViT). ICLR 2021.